EXPERIMENT 6: AUTOMATING BUSINESS PROCESSES USING DELUGE SCRIPTING AND WORKFLOWS, WITH APPLICATION TESTING AND DEBUGGING

Aim

To automate business processes in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System using Deluge scripting and workflows, and to perform application testing and debugging to ensure accurate workflow execution, data validation, and recruitment process automation.

Theory

In Zoho Creator, business process automation is built using Workflows and Deluge (Data Enriched Language for Universal Application Environment) scripting. Workflows allow developers to implement event-driven automation, enforce strict business constraints, automate multi-channel communications, and streamline end-to-end recruitment procedures without traditional complex backend development.

Key concepts in workflow automation, testing, and debugging include:

- **Workflow Triggers & Event Handlers:** Workflows execute automatically in response to key application events:
  • On Load: Pre-populates default form values, initializes current dates, and sets initial record states.
  • On User Input: Executes real-time dynamic validations, field masking, duplicate checking, and conditional display of fields as the user fills out a form.
  • On Success: Runs post-submission tasks such as automated database updates, multi-form record synchronization, and email notifications.
  • Scheduled Workflows: Time-based automated routines running at specified intervals (e.g., daily schedule) to perform batch operations and reminders.
- **Deluge Scripting Engine:** Deluge is an integrated low-code scripting language in Zoho Creator used to execute custom business logic, run database queries (Fetch, Insert, Update, Delete), perform string/date manipulations, implement conditional branching (if-else statements), and invoke built-in functions like sendmail.
- **Data Validation & Business Logic:** Deluge scripts enforce critical business rules prior to record commit (e.g., preventing past interview date selection, verifying candidate email uniqueness, or validating scoring ranges), raising clear user alerts and halting invalid form submissions using the cancel submit command.
- **Automated Notifications:** Integrated notification systems automatically notify candidates, interviewers, and HR managers at key milestone events (e.g., application acknowledgment, interview invitation, selection/rejection status updates).
- **Application Testing & Debugging:** Rigorous testing involves creating test candidate records, scheduling mock interviews, and simulating status transitions across edge cases. Debugging relies on

Zoho Creator's Execution Logs (Executed Script Logs), info statements for real-time variable tracking, and error diagnostic tools to identify and resolve logical or syntax errors.

Procedure

1. Log in to Zoho Creator and open the existing Automated Corporate Talent Acquisition and Interview Tracking System application.

2. Navigate to the Workflows section from the application editing workspace.

3. Create an On Load workflow on the Candidate Application form to set default values, such as setting the Application Status to 'New Application' and populating the current submission date.

4. Create an On User Input workflow on the Candidate Email field to check for duplicate applicant records using a Deluge search query, displaying an alert message if a matching email exists.

5. Configure an On User Input validation workflow on the Interview Scheduling form to verify that scheduled interview dates and times are set in the future, canceling submission if an invalid date is entered.

6. Create an On Success workflow on the Candidate Application form using Deluge scripting (sendmail) to automatically send a confirmation email to the candidate upon successful submission.

7. Configure an On Success workflow on the Interview Evaluation form that automatically updates the candidate's recruitment status (e.g., 'Shortlisted', 'Selected', or 'Rejected') in the primary candidate record based on evaluation scores.

8. Set up a Scheduled Workflow running daily to check for upcoming interviews and dispatch automated reminder notifications to recruiters and candidates 24 hours prior to scheduled interview times.

9. Perform Application Testing by submitting test candidate records, scheduling sample interviews, and triggering candidate status transitions across various scenarios.

10. Open the Logs / Executed Script Logs console in Zoho Creator and use info statements to trace variable values, inspect workflow execution paths, and verify that scripts complete without errors.

11. Debug and resolve any logical errors, missing fields, or failed email tasks encountered during testing to ensure smooth end-to-end recruitment process execution.

Sample Deluge Scripts & Automation Logic

Below are reference Deluge code implementations configured and tested during the experiment:

1. On Load Workflow: Setting Defaults (Candidate Form)

```
// On Load Event - Candidate Application Form
input.Application_Status = "New Application";
input.Application_Date = zoho.currentdate;
info "Form initialized with status: New Application and date: " +
input.Application_Date;
```

2. On User Input Workflow: Email Uniqueness & Validation

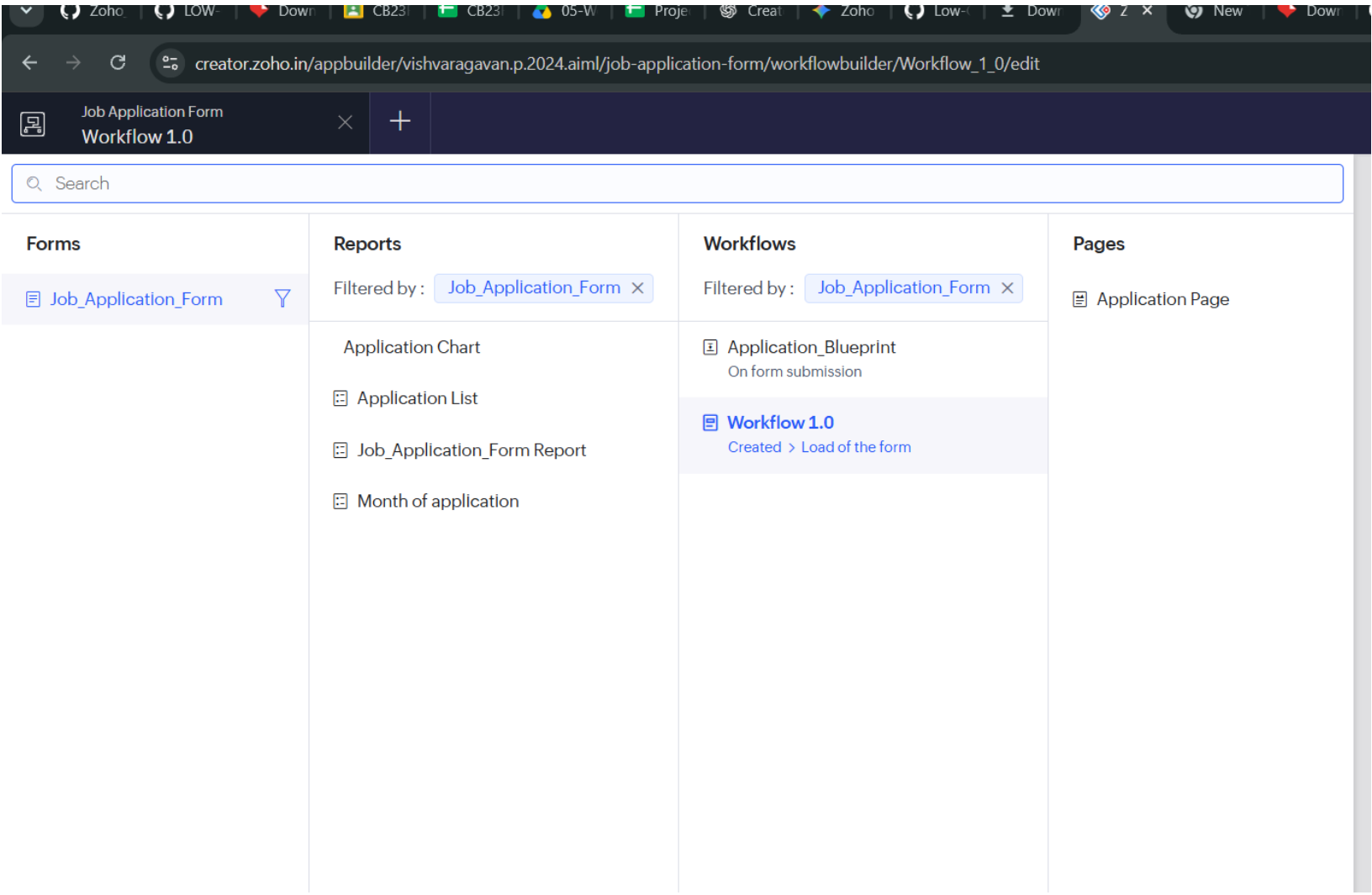
```
// On User Input Event - Candidate Email Field
existingCount = Candidate_Application[Email == input.Email].count();
if (existingCount > 0)
{
    alert "A candidate application with the email address '" + input.Email + "'
already exists.";
    input.Email = "";
}
```

3. On User Input Workflow: Future Interview Date Validation

```
// On User Input / Validation Event - Interview Scheduling Form
if (input.Interview_Date < zoho.currentdate)
{
    alert "Interview Date cannot be in the past. Please select a valid future
date.";
    cancel submit;
}
```

4. On Success Workflow: Automatic Email Acknowledgment & Status Update

```
// On Success Event - Candidate Application Form
sendmail
[
    From    : zoho.adminuserid
    To      : input.Email
    Subject : "Application Received - Corporate Talent Acquisition"
    Message : "Dear " + input.First_Name + ",<br><br>Thank you for applying for the
position of " + input.Applied_Position + ". Your application ID is " + input.ID + ".
We will review your profile and contact you soon.<br><br>Best regards,<br>Talent
Acquisition Team"
];
info "Confirmation email successfully sent to: " + input.Email;
```


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/workflowbuilder/Workflow_1_0/edit
Job Application Form
Workflow 1.0
Search
Forms
Job_Application_Form
Reports
Filtered by : Job_Application_Form
Application Chart
Application List
Job_Application_Form Report
Month of application
Workflows
Filtered by : Job_Application_Form
Application_Blueprint
On form submission
Workflow 1.0
Created > Load of the form
Pages
Application Page

Job Application Form
Workflow 1.0

Job_Application_Form › Created › Load of the form › **Workflow 1.0**

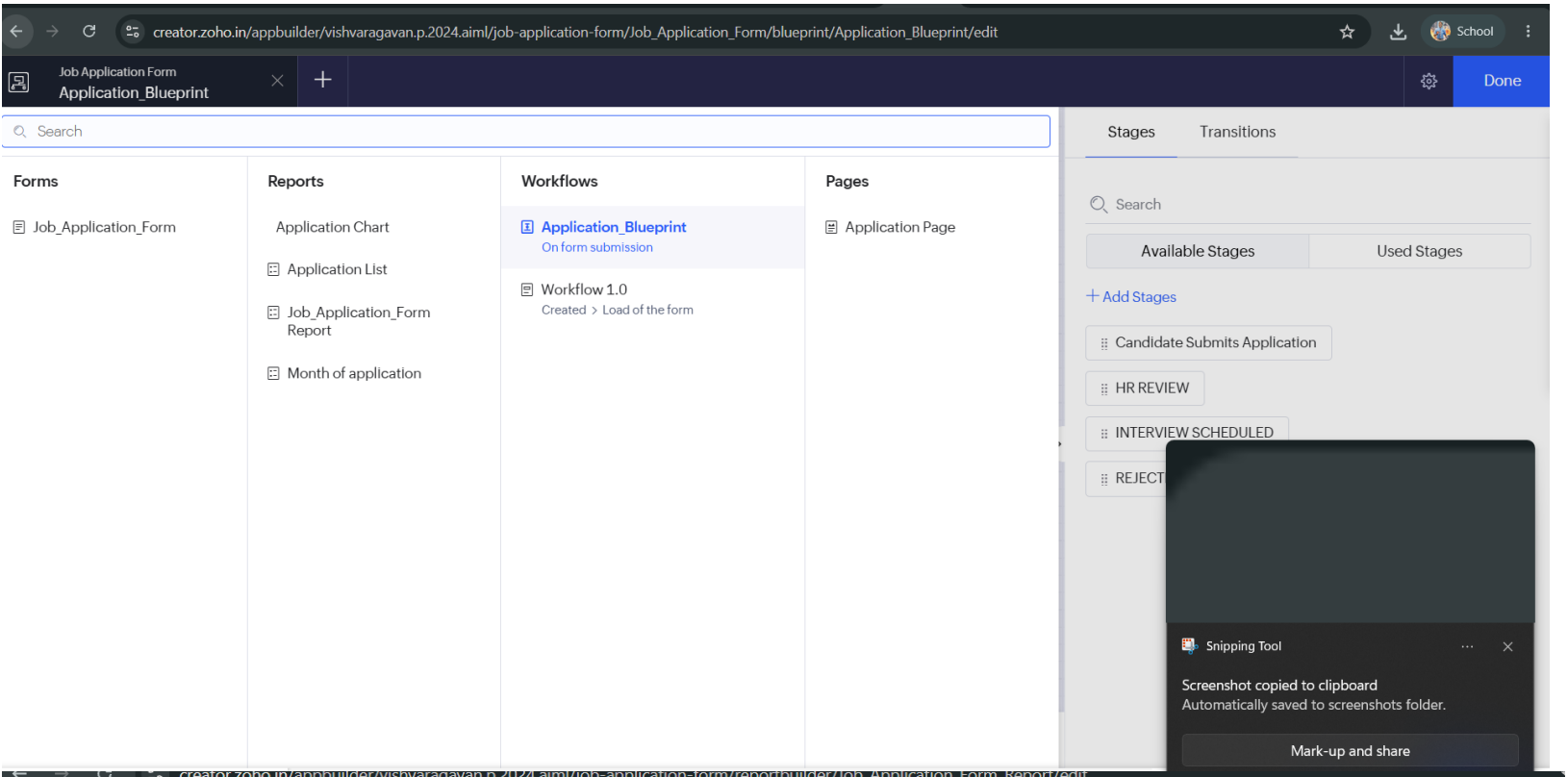
Actions will be executed when the form loads. Learn more

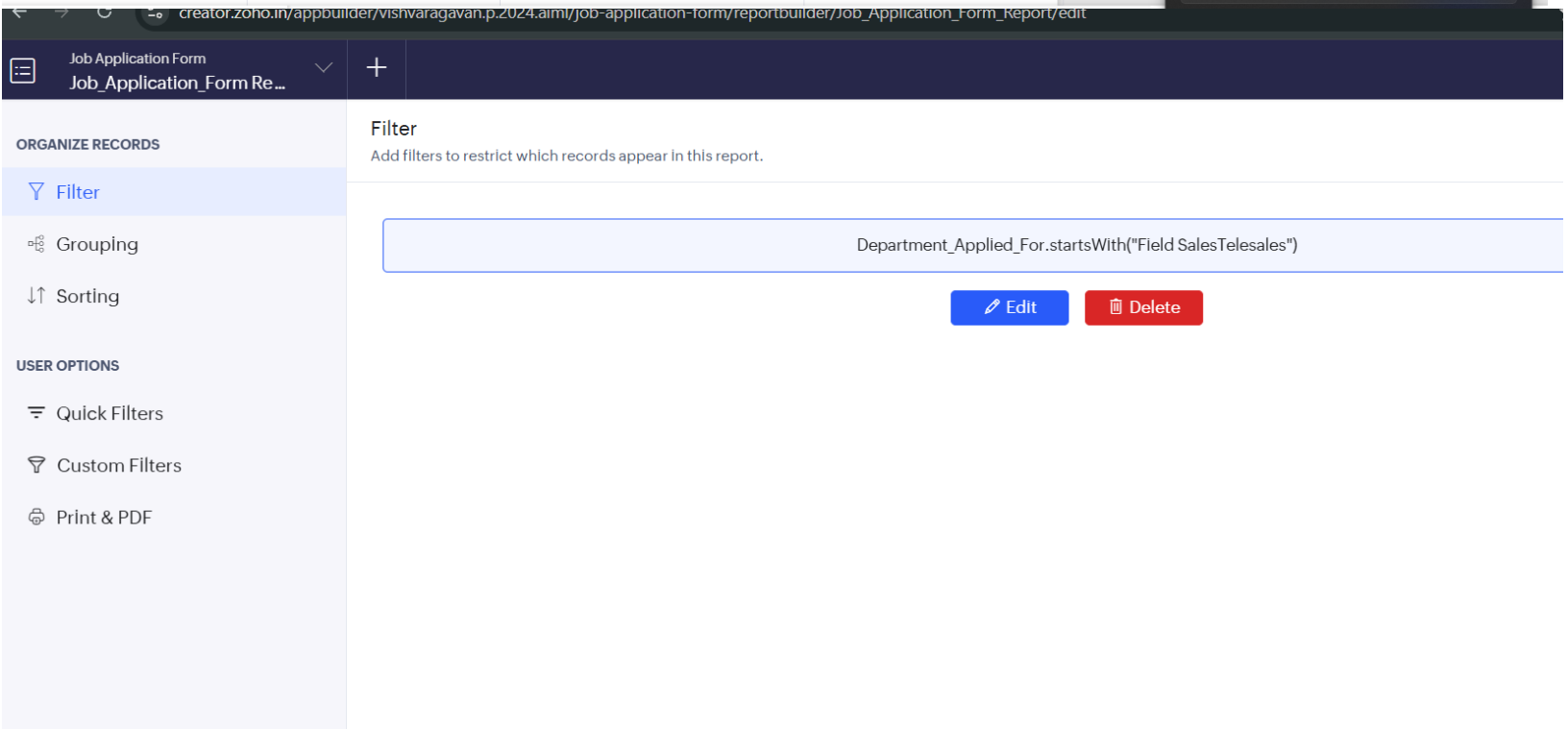
Execute without condition

fx Deluge script

```
1  //hello
2  input.Department_Applied_For = "Software Development";
3  input.Years_of_Experience = 0;
4  input.I_consent_to_background_verification_if_selected = {"Choice 1"};
5  input.Available_Joining_Date = zoho.currentdate;
6  input.Preferred_Interview_Time = zoho.currenttime;
7  info "Welcome to Job Application Form";
8
```

Add New Action


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/Job_Application_Form/blueprint/Application_Blueprint/edit
School
Job Application Form
Application_Blueprint
Done
Search
Forms
Job_Application_Form
Reports
Application Chart
Application List
Job_Application_Form Report
Month of application
Workflows
Application_Blueprint
On form submission
Workflow 1.0
Created > Load of the form
Pages
Application Page
Stages
Transitions
Search
Available Stages
Used Stages
Add Stages
Candidate Submits Application
HR REVIEW
INTERVIEW SCHEDULED
REJECT
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/reportbuilder/Job_Application_Form_Report/edit
Job Application Form
Job_Application_Form Re...
ORGANIZE RECORDS
Filter
Grouping
Sorting
USER OPTIONS
Quick Filters
Custom Filters
Print & PDF
Filter
Add filters to restrict which records appear in this report.
Department_Applied_For.startsWith("Field SalesTelesales")
Edit
Delete

Result

Thus, automated business processes were implemented using Deluge scripting and workflows in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System. The application was thoroughly tested and debugged to verify accurate workflow triggers, field validations, automated notifications, and recruitment status tracking for recruiters and hiring management.